How to startup the AKD2G Motor controller

1. **Connect power**
    a. Connect 230V to X3:
        i. Yellow on PE
        ii. Brown on L1
        iii. Blue on L2
    b. Connect 24VDC to X10
2. **Connect e-motors**
    a. Connect motor 1 on X1
    b. Connect motor 2 on X2
3. **Connect Service cable**
    a. Connect a RJ45-cable to X20 and on the other side to the computer
4. **Connect IO-output with 24VDC/0VDC for motor 1**

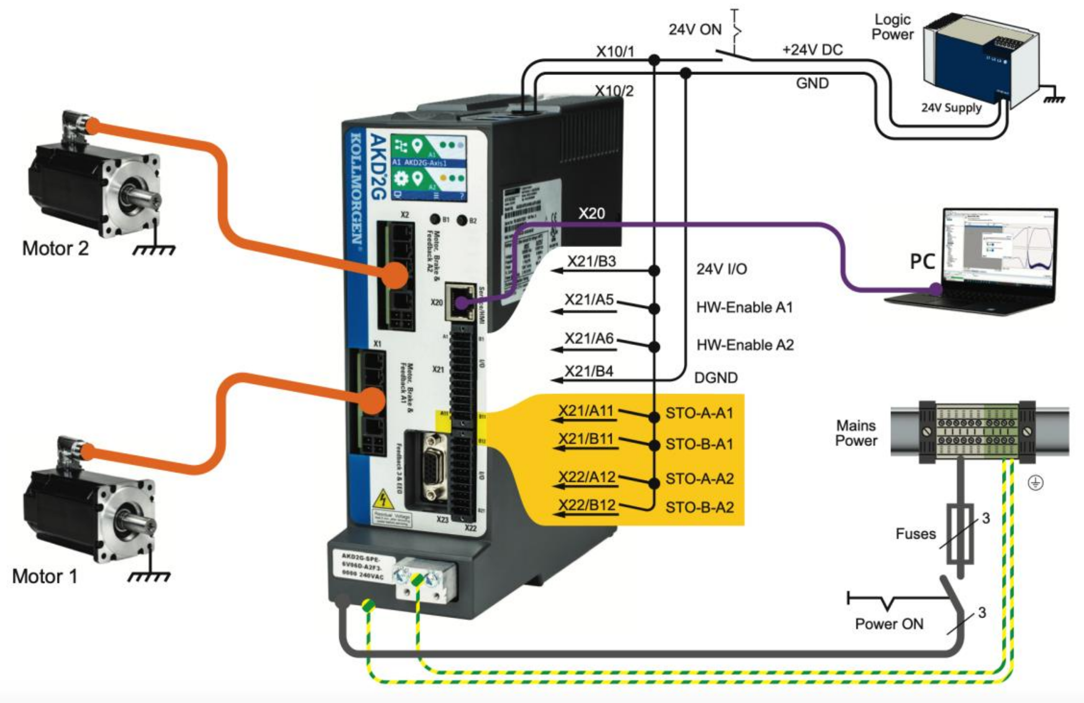
JUST CONNECT THE TWO BLACK SWITCHES TO THE ESC AND FOLLOW THE CABLES; THERE SHOULD BE JUST TWO FREE, POSITIVE AND NEGATIVE, CONNECT BOTH TO THE POWER SUPPLY. PLUG AND PLAY

a. X21-B3 24VDC
b. X21-B4 0VDC
c. X21-A11 24VDC
d. X21-B11 24VDC

e. X21-A5 24VDC

9.2.1.2 Minimum wiring for drive test without load, example

NOTICE This wiring diagram based on default settings is for general illustration only and does not fulfill any requirements for EMC, functional safety, or functionality of your application.

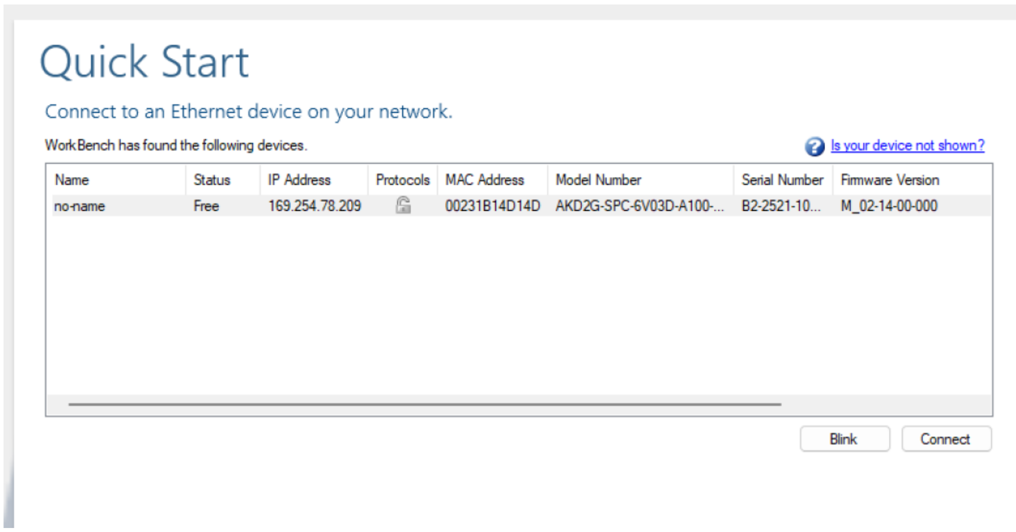


Installation and Set-up

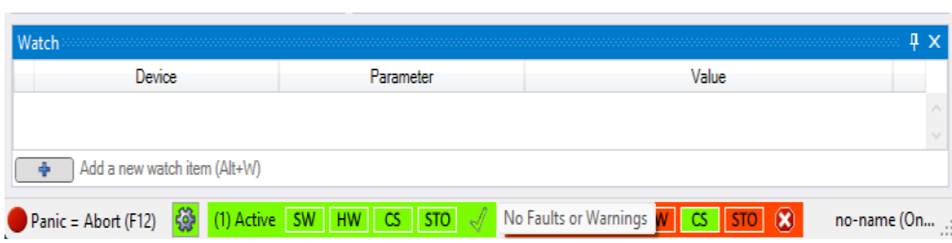
1. Go to website : https://www.kollmorgen.com/en-us/developer-network/akd2g-downloads
2. Download Safe motion monitor v2 (SMM2) *Safety Certified *

Title	Release Information	Older Versions
Safe Motion Monitor v1 (SMM1) Package * Safety Certified *	1.04	view all
Safe Motion Monitor v2 (SMM2) Package * Safety Certified *	2	view all
AKD2G with FS2/FS3 Servo Drive Firmware * Safety Certified *	02-12-00-105	view all
AKD2G Firmware for KAS	02-14-00-000	view all
AKD2G with FS1 Servo Drive Firmware	02.14.00.000	view all
AKD2G Firmware Release Notes EN		view all

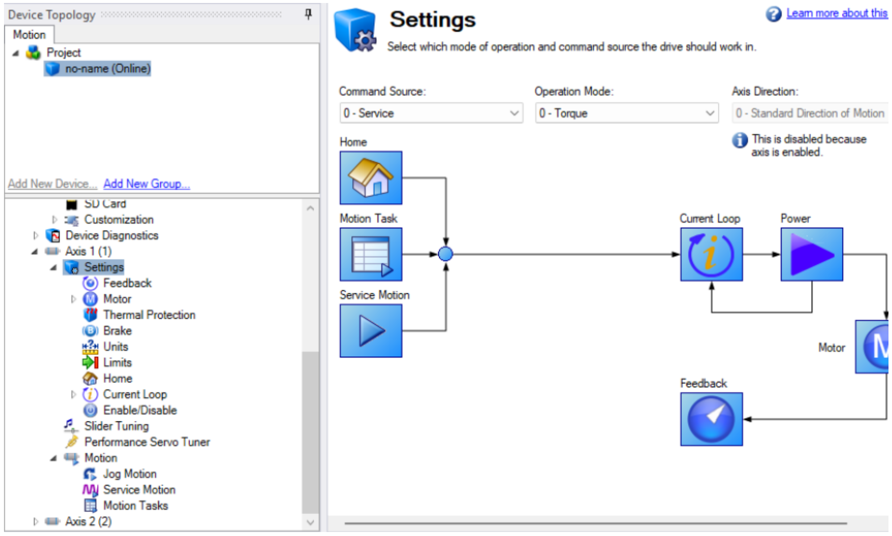
3. After Downloading, open folder Workbench and run Kollmorgen WorkBench setup and install.
4. Open Workbench and connect the HMI cable to your pc. The device will show up as following



5. Click on connect. New Window should pop-up
6. On the left pane, go to device settings>power> AC line Phases. Set to 0- Single Phase.
7. Then on the left pane, Click on Axis 1 in axis overview set axis status as enable, (For Axis to be enabled STO must be off which is done with wiring 24v supply to A11 and B11 as shown in the connection description on page 1)
8. If there are no errors a green bar with a tick should show up on the bottom of the screen.



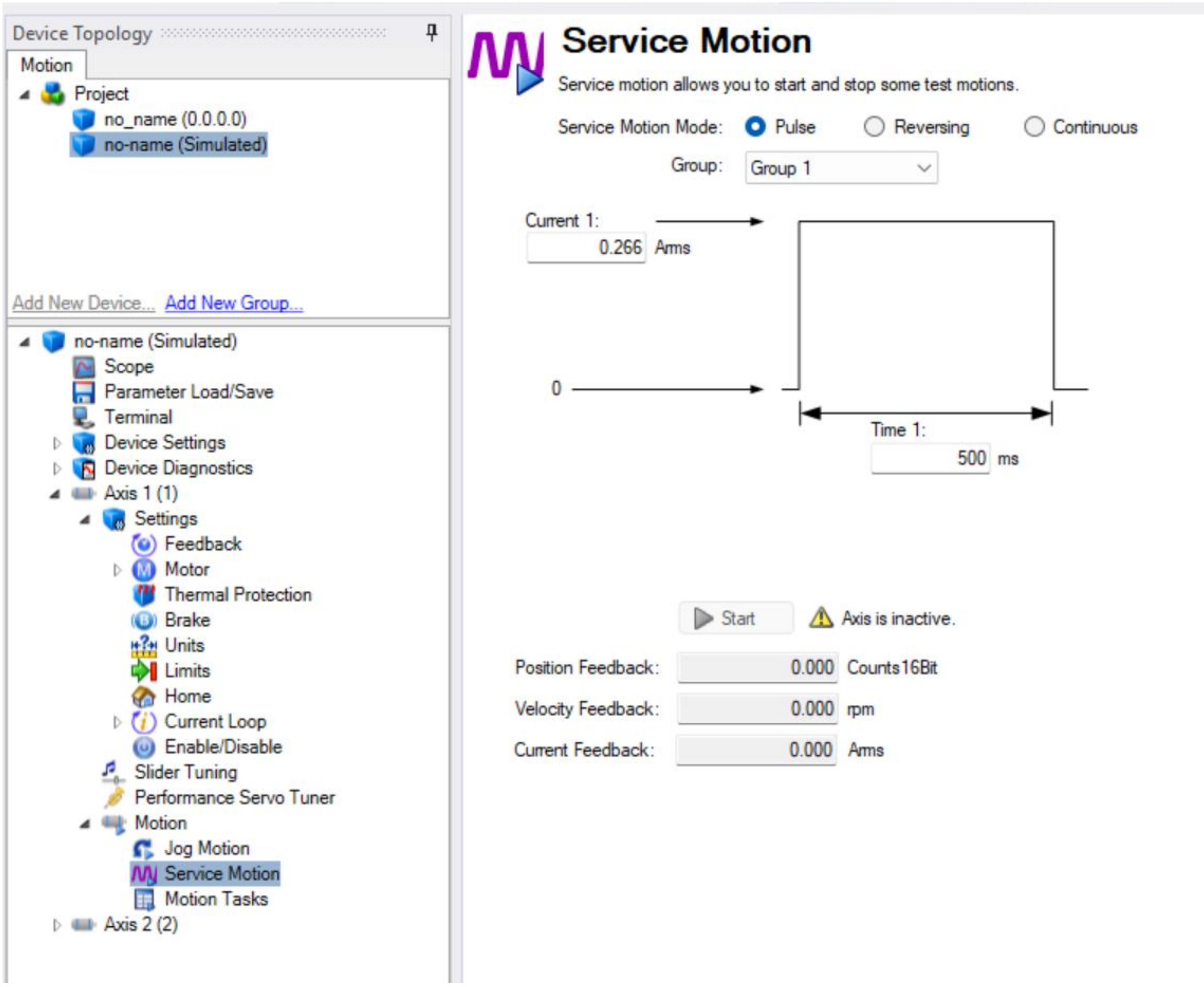
9. To test the motor, expand Axis 1> Motion> Service motion. Set current at 1 Amps and click on start.
10. Default Operation mode is torque which can be changed under Axis 1>Settings.

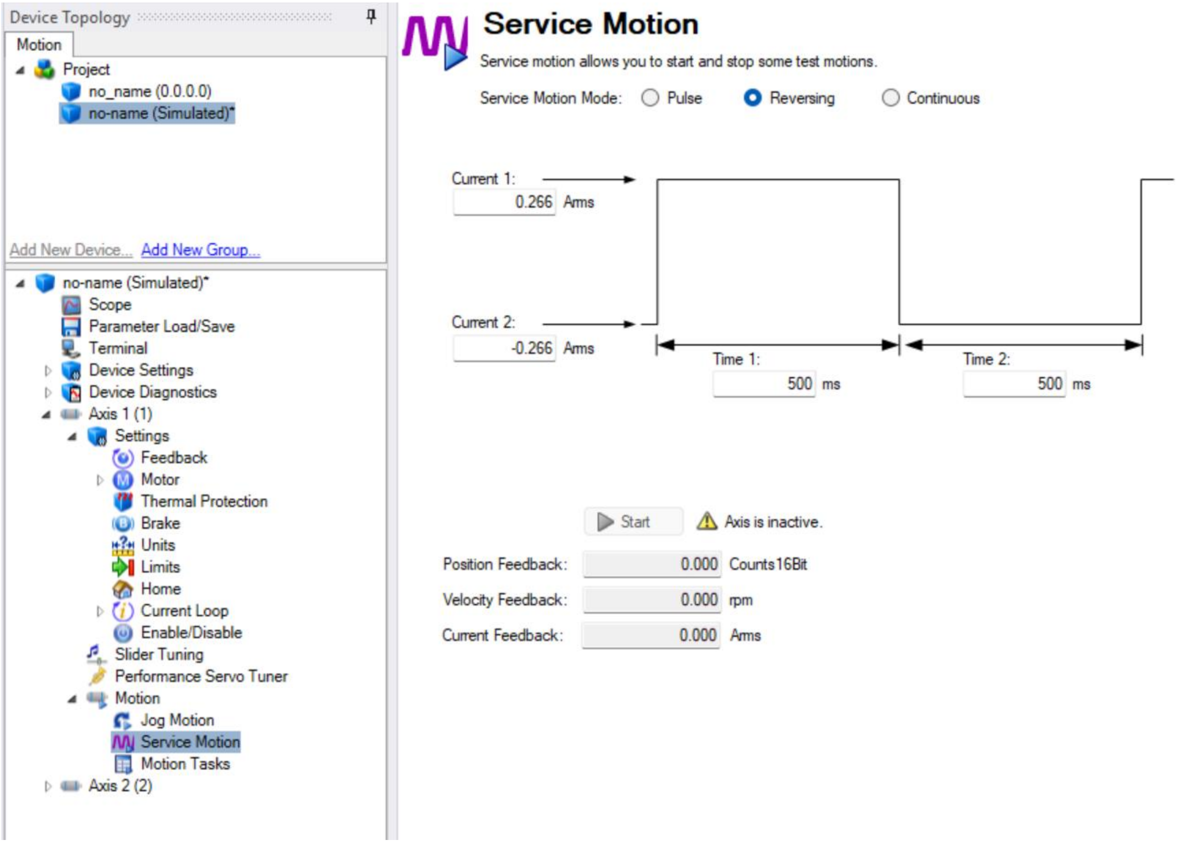


11. Once the motors are connected, following the previous steps, go to the Service Motions option. There are three options to move the motors manually, as shown in the images. The movement speed is determined by the amount of current (Amperes): with positive current the arm extends, and with negative current the arm retracts.

    The three options are: a single pulse with the duration set in milliseconds, another option with two pulses sent periodically, and a third option with continuous current. There is a movement limit, after which it brakes and no longer moves.

    WARNING: WE DO NOT KNOW IF THERE IS CURRENT LIMITATION (SPEED OF MOVEMENT) BASED ON THE FREEDOM OF MOTION OF THE GIMBAL. EXCESSIVE CURRENT COULD MECHANICALLY DAMAGE THE ARMS.


Device Topology
Motion
Project
no_name (0.0.0.0)
no-name (Simulated)
Add New Device...
Add New Group...
no-name (Simulated)
Scope
Parameter Load/Save
Terminal
Device Settings
Device Diagnostics
Axis 1 (1)
Settings
Feedback
Motor
Thermal Protection
Brake
Units
Limits
Home
Current Loop
Enable/Disable
Slider Tuning
Performance Servo Tuner
Motion
Jog Motion
Service Motion
Motion Tasks
Axis 2 (2)
Service Motion
Service motion allows you to start and stop some test motions.
Service Motion Mode:
Pulse
Reversing
Continuous
Group:
Group 1
Current 1:
0.266
Arms
0
Time 1:
500
ms
Start
Axis is inactive.
Position Feedback:
0.000
Counts16Bit
Velocity Feedback:
0.000
rpm
Current Feedback:
0.000
Arms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Arms
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ms
Time 2:
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rpm
Current Feedback:
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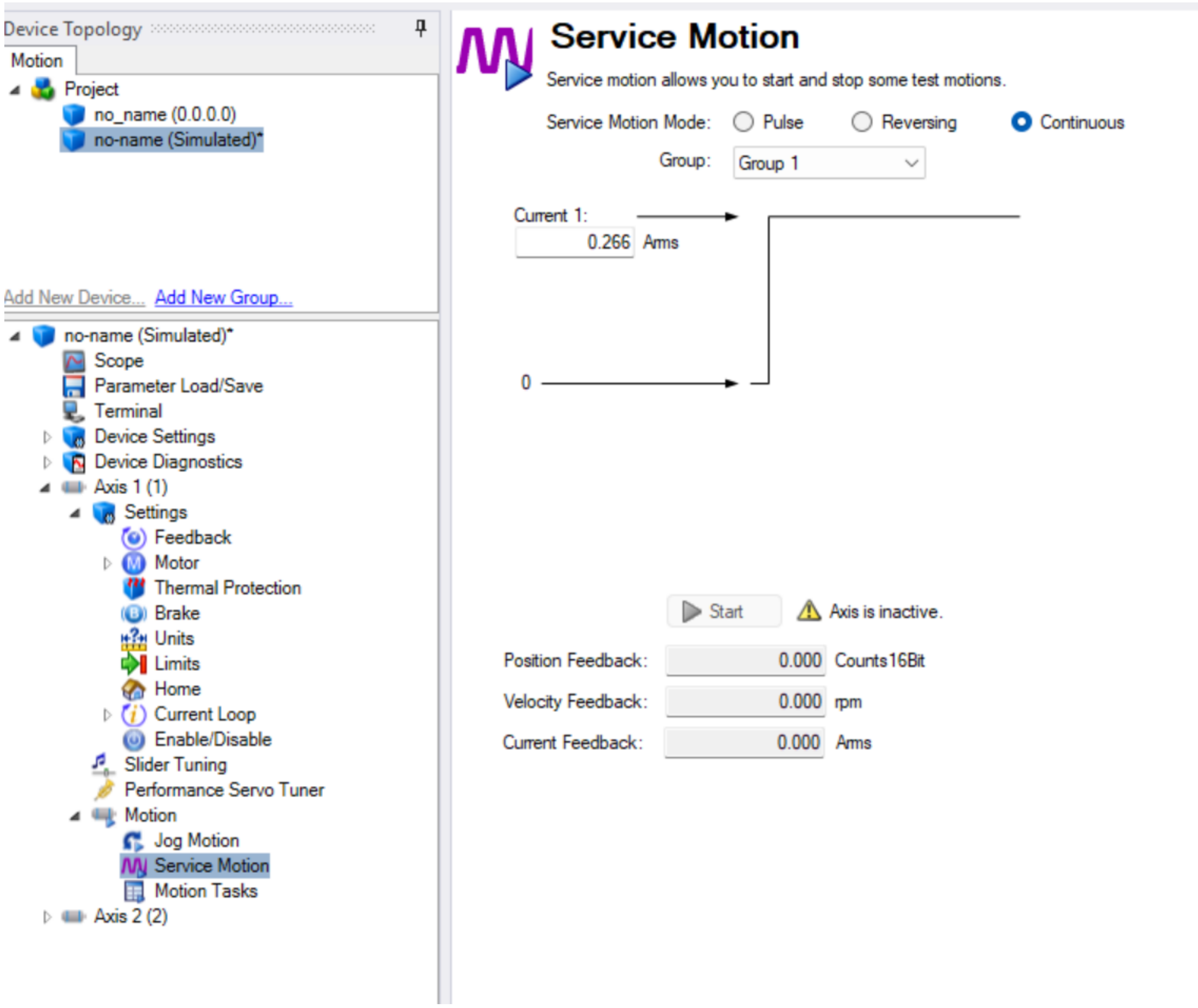
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